

Of course! I've extracted all the formulas and tables from the document. Here they are organized in Markdown format:

Formulas

2.2.1 Solar Energy Transmittance of Glazing

$$g_{gl;wt} = F_w \cdot g_{gl;n}$$

Where:

- $g_{gl;wt}$ = Total solar energy transmittance of glazing
- F_w = Correction factor for glass dispersion (0.90)
- $g_{gl;n}$ = Total solar energy transmittance at normal incidence

2.2.3 Monthly Average Solar Energy Transmittance with Movable Shading

$$g_{gl;wt;m} = (1 - f_{sh;wt;m}) \cdot g_{gl;wt} + f_{sh;wt;m} \cdot g_{gl;sh;wt}$$

Where:

- $g_{gl;wt;m}$ = Monthly average total solar energy transmittance
- $f_{sh;wt;m}$ = Fraction of time with movable shading device activated
- $g_{gl;wt}$ = Total solar energy transmittance without shading
- $g_{gl;sh;wt}$ = Total solar energy transmittance with shading activated

Tables

Table 11: Solar Energy Transmittance for Different Glass Types

Type	$g_{gl;n}$	$g_{gl;wt}$
Single glass	0.85	0.77

Type	g _{gl;n}	g _{gl;wt}
Double glass	0.75	0.68
Double low-emissivity glass	0.67	0.60
Triple low-emissivity glass	0.50	0.45
Double window	0.75	0.68

Table 12: Solar Energy Transmittance with Movable Shading Devices (g_{d;sh;wl})

(This is a complex table showing values for different glass types, shading types, positions, and colors - too extensive to reproduce here fully)

Structure:

- Shading types: Exterior vs Interior protection
- Solar transmittance factors (τ_{0,B}): 0 (e.g., blinds), 0.2 (e.g., awnings), 0.4 (e.g., curtains)
- Colors: white, pastel, dark, black
- Glass types: single, double, double low-e, triple low-e

Table 13: Solar Protection Effectiveness

Class (UNE-EN 14501)	0	1	2	3	4
Effectiveness	Minimum effect	Small effect	Moderate effect	Efficient	Very efficient
g _{d;sh;wl}	0.5	< 0.5	< 0.35	< 0.15	< 0.10

Table 14: Operation Setpoints for Movable Shading Devices

Device Position	Manual or motorized with manual control	Motorized with automated control
Open (disconnected or inactive)	I _{sol} < 300 W/m²	I _{sol} < 200 W/m²

Device Position	Manual or motorized with manual control	Motorized with automated control
Closed (connected or active)	$I_{sol} > 300 \text{ W/m}^2$	$I_{sol} > 200 \text{ W/m}^2$

Tables 15.a & 15.b: Activation Fractions by Month and Orientation

(Extensive tables showing monthly activation fractions $f_{sh;wt}$ for different climate zones and orientations)

Coverage:

- Table 15.a: Peninsular Spain, Balearic Islands, Ceuta and Melilla
- Table 15.b: Canary Islands
- Climate zones: A4, B4, C3, C4, D1, D2, D3, E1, etc.
- Orientations: E (East), S (South), O (West), N (North)
- Monthly values from January to December

Table 16: Shading Factor for Overhangs ($F_{sh;obst}$)

Orientation	D/H Range	L/H: 0.2-0.5	L/H: 0.5-1	L/H: 1-2	L/H: >2
S	$0 < D/H \leq 0.2$	0.82	0.50	0.28	0.16
S	$0.2 < D/H \leq 0.5$	0.87	0.64	0.39	0.22
S	$D/H > 0.5$	0.93	0.82	0.60	0.39
SE/SO	$0 < D/H \leq 0.2$	0.90	0.71	0.43	0.16
SE/SO	$0.2 < D/H \leq 0.5$	0.94	0.82	0.60	0.27
SE/SO	$D/H > 0.5$	0.98	0.93	0.84	0.65
E/O	$0 < D/H \leq 0.2$	0.92	0.77	0.55	0.22
E/O	$0.2 < D/H \leq 0.5$	0.96	0.86	0.70	0.43
E/O	$D/H > 0.5$	0.99	0.96	0.89	0.75

Table 17: Shading Factor for Recesses ($F_{sh;obst}$)

Orientation	R/H Range	R/W: 0.05-0.1	R/W: 0.1-0.2	R/W: 0.2-0.5	R/W: >0.5
S	$0.05 < R/H \leq 0.1$	0.82	0.74	0.62	0.39
S	$0.1 < R/H \leq 0.2$	0.76	0.67	0.56	0.35
S	$0.2 < R/H \leq 0.5$	0.56	0.51	0.39	0.27
S	$R/H > 0.5$	0.35	0.32	0.27	0.17
SE/SO	$0.05 < R/H \leq 0.1$	0.86	0.81	0.72	0.51
SE/SO	$0.1 < R/H \leq 0.2$	0.79	0.74	0.66	0.47
SE/SO	$0.2 < R/H \leq 0.5$	0.59	0.56	0.47	0.36
SE/SO	$R/H > 0.5$	0.38	0.36	0.32	0.23
E/O	$0.05 < R/H \leq 0.1$	0.91	0.87	0.81	0.65
E/O	$0.1 < R/H \leq 0.2$	0.86	0.82	0.76	0.61
E/O	$0.2 < R/H \leq 0.5$	0.71	0.68	0.61	0.51
E/O	$R/H > 0.5$	0.53	0.51	0.48	0.39

Table 18: Shading Factor for Louvers ($F_{sh;obst}$)

Horizontal Louvers: | Orientation | Tilt Angle: 0° | 30° | 60° | |-----|-----|----|----| | South
| 0.49 | 0.42 | 0.26 | | Southeast/Southwest | 0.54 | 0.44 | 0.26 | | East/West | 0.57 | 0.45 | 0.27 |

Vertical Louvers: | Orientation | -60° | -45° | -30° | 0° | 30° | 45° | 60° | |-----|-----|----|----|----|
----|----|----| | South | 0.37 | 0.44 | 0.49 | 0.53 | 0.47 | 0.41 | 0.32 | | Southeast | 0.46 | 0.53 | 0.56 |
0.56 | 0.47 | 0.40 | 0.30 | | East | 0.39 | 0.47 | 0.54 | 0.63 | 0.55 | 0.45 | 0.32 | | West | 0.44 | 0.52 |
0.58 | 0.63 | 0.50 | 0.41 | 0.29 | | Southwest | 0.38 | 0.44 | 0.50 | 0.56 | 0.53 | 0.48 | 0.38 |

Table 19: Shading Factor for Skylights ($F_{sh;obst}$)

(Matrix table with Y/Z ratios from 0.1 to 10.0)

Key values:

- Minimum: 0.42 ($Y/Z = 0.1/0.1$)
- Maximum: 0.85 ($Y/Z = 10.0/10.0$)

Key Parameters Definition

Geometric Parameters:

- L = Overhang projection length
- H = Window height
- D = Vertical distance from window top to overhang
- R = Recess depth
- W = Window width
- Y/Z = Ratio for skylights (dimension characteristics)

Orientation Codes:

- S = South ($162^\circ \leq \alpha_0 < 198^\circ$)
- SE = Southeast ($111^\circ \leq \alpha_0 < 162^\circ$)
- E = East ($60^\circ \leq \alpha_0 < 111^\circ$)
- O = West ($249^\circ \leq \alpha_0 < 300^\circ$)
- SO = Southwest ($198^\circ \leq \alpha_0 < 249^\circ$)
- N = North ($\alpha_0 < 22.5^\circ$ or $\alpha_0 \geq 337.5^\circ$)

This document provides the complete methodology for calculating solar shading factors that I mentioned in my previous answer about obstruction factors for fenestration.